

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	C	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,C	B,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	B	C	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	2	5	3	2	60	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	D	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	C,D	A,B,C	A,B,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	B	D	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	4	12	85	20	1140	39

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	A	A	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C	A,C	B,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	B	D	A	D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	1	6	9	31	8	23

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (D)**

$$\text{Acceleration } a = \frac{dv}{dt}$$

$$\Rightarrow a = 2bt$$

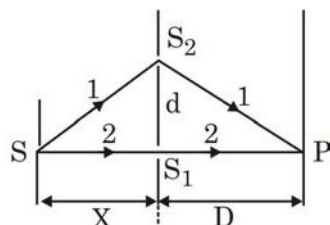
For block to just slide on the plate, we have

$$\mu mg = m(2bt)$$

$$\Rightarrow t = \frac{\mu g}{2b}$$

2. Ans (A)

Refer to figure, to reach point P, wave 1 has to travel a path ($SS_2 + S_2P$) while wave 2 has to travel a path ($SS_1 + S_1P$). Therefore, when the waves arrive at P, the path difference is



$$\delta = (SS_2 + S_2P) - (SS_1 + S_1P)$$

Now, in triangle SS_2S_1 , we have(1)

$$SS_2 = (x^2 + d^2)^{1/2} = \left(1 + \frac{d^2}{x^2}\right)^{1/2}$$

$$= \left(1 + \frac{d^2}{2x^2}\right) (\because d \ll x)$$

$$\text{Similarly, } S_2P = (D^2 + d^2)^{1/2} = D \left(1 + \frac{d^2}{2D^2}\right)$$

$$(\because d \ll D)$$

Also $(SS_1 + S_1P) = x + D$, Using these in Eq.

(1), we have

$$\Delta = x \left(1 + \frac{d^2}{2x^2}\right) + D \left(1 + \frac{d^2}{2D^2}\right) - (x + D)$$

$$= x + \frac{d^2}{2x} + D + \frac{d^2}{2D} - x - D$$

$$\text{or } \Delta = \frac{d^2}{2} \left(\frac{1}{x} + \frac{1}{D}\right)$$

In order to have a dark fringe at P, $\Delta = \frac{\lambda}{2}$.

Hence

$$\frac{\lambda}{2} = \frac{d^2}{2} \left(\frac{1}{x} + \frac{1}{D}\right) \text{(1)}$$

$$\text{or } d = \left[\frac{\lambda x D}{x + D}\right]^{1/2} \text{(2)}$$

Putting $x = \frac{D}{2}$ in Eq. (2), we find that the correct option is (A).

4. Ans (A)

Maximum speed is attained when the entire energy stored in inductor is converted into kinetic energy of rod.

$$\Rightarrow \frac{1}{2} L \left(\frac{E}{R}\right)^2 = \frac{1}{2} m v^2$$

$$\Rightarrow v = \left(\frac{E}{R}\right) \sqrt{\frac{L}{m}}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C,D)

$$\Rightarrow dI = kr^2 (2\pi r dr)$$

$$\Rightarrow I = 2\pi k \int_0^a r^3 dr$$

$$\Rightarrow I = \frac{1}{2} \pi k a^4$$

Further, field for $r > a$ is

$$B = \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 k a^4}{4r}$$

and field for $r < a$ is

$$B = \frac{\mu_0 \left(\frac{\pi k r^4}{2}\right)}{2\pi r} = \frac{1}{4} \mu_0 k r^3$$

6. Ans (A,C)

When number of electrons increases, then number of photons emitted also increases, thereby increasing the intensity. Energy, wavelength and frequency of photon are related to potential difference.

7. Ans (B,C,D)

$$\text{Work done by } \vec{F}_1 \text{ is } W_1 = \int_{P_1}^{P_2} F_1 \cos \theta ds$$

Here, $ds = (6)d(-2\theta) = -12 d\theta$ and $F_1 = 20 \text{ N}$

$$\therefore W_1 = -240 \int_{\pi/4}^0 \cos \theta d\theta = 240 \sin \frac{\pi}{4} = 120 \sqrt{2} \text{ J}$$

$\vec{F}_1$ is conservative because it is always directed towards a fixed point P_2 .

$$W_2 = F_2(OP_2) = (30)(6) = 180 \text{ J and}$$

$$W_3 = \vec{F}_1 \int_0^{6(\pi/2)} F_3 ds = \int_0^{3\pi} 15 ds$$

$$= [15x]_0^{3\pi} = 45\pi$$

PART-1 : PHYSICS

SECTION-I (iii)

11. Ans (B)

disc gives uniform E so force on dipole is zero.

Electric field inside a sheet varies linearly.

PART-1 : PHYSICS

SECTION-II

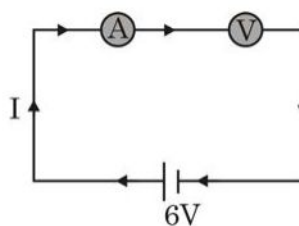
1. Ans (2)

Let R_1 = resistance of ammeter and R_2 = combined resistance of ammeter and voltmeter

In the first case, current in the circuit is given by

$$I = \frac{6}{R_2} \dots\dots\dots(1)$$

and voltage across voltmeter is given by



$$V = 6 - (\text{voltage across ammeter})$$

$$\Rightarrow V = 6 - IR_1$$

$$\Rightarrow V = 6 - \frac{6R_1}{R_2} \dots\dots\dots(2)$$

In the second case reading of ammeter becomes two times, i.e., the total resistance becomes half whereas the resistance of ammeter remains unchanged.

Hence,

$$I' = \frac{6}{R_2/2} = \frac{12}{R_2} \dots\dots\dots(3)$$

$$\text{and } V' = 6 - (I')R_1$$

$$\Rightarrow V' = 6 - \frac{12R_1}{R_2} \dots\dots\dots(4)$$

$$\text{Further, it is given that } V' = \frac{V}{2}$$

$$\Rightarrow 6 - \frac{12R_1}{R_2} = 3 - \frac{3R_1}{R_2}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{1}{3}$$

Substituting this value in equation (4), we have

$$V' = 6 - (12) \left(\frac{1}{3} \right)$$

$$\Rightarrow V' = 2V$$

2. Ans (5)

For an adiabatic process $TV^{\gamma-1} = \text{constant}$

$$\Rightarrow TV^{\gamma-1} = \left(\frac{T}{2}\right) (5.66V)^{\gamma-1}$$

$$\Rightarrow (5.66V)^{\gamma-1} = 2$$

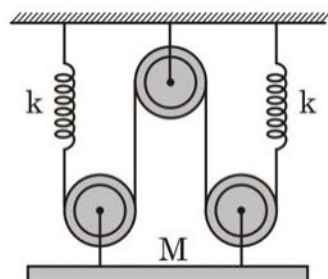
$$\Rightarrow (\gamma - 1)\log_e(5.66) = \log_e(2)$$

$$\Rightarrow \gamma - 1 = 0.4$$

$$\Rightarrow \gamma = 1.4$$

$$\text{Since, } \gamma = 1 + \frac{2}{f} = 1.4$$

4. Ans (2)



Net restoring force is

$$F = -8x$$

$$\Rightarrow Ma = -8kx$$

$$\Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{a}{x}}$$

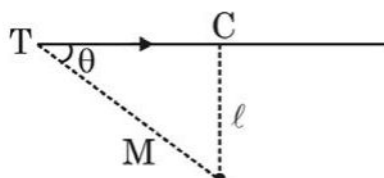
$$\Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{8k}{M}} = \frac{1}{\pi} \sqrt{\frac{2k}{M}}$$

$$\Rightarrow x = 2$$

5. Ans (60)

Let the velocity of truck at T when it blows the whistle be v_s .

Then



$$600 = \left(\frac{v}{v - v_s \cos \theta} \right) 500 \dots\dots\dots(1)$$

During this time, speed of truck gets doubled, so

$$2v_s = v_s + at$$

$$\Rightarrow v_s = at \dots\dots\dots(2)$$

$$\text{Now, } vt = AM = \frac{\ell}{\sin \theta} \dots\dots\dots(3)$$

$$\text{Also, } AB = \ell \cot \theta = v_s t + \frac{1}{2} at^2 \dots\dots\dots(4)$$

$$\Rightarrow \frac{\ell \cos \theta}{\sin \theta} = at^2 + \frac{1}{2} at^2 = \frac{3}{2} at^2$$

$$\Rightarrow \frac{a\ell}{v} = \left(\frac{2}{3} \sin \theta \cos \theta \right) v$$

From equation (1), we get

$$\frac{6}{5} = \frac{u}{v - at \cos \theta} = \frac{v}{v - \frac{a\ell}{v} \left(\frac{\cos \theta}{\sin \theta} \right)}$$

$$\left\{ \because t = \frac{\ell}{v \sin \theta} \right\}$$

$$\Rightarrow \frac{6}{5} = \frac{v}{v - \left(\frac{2}{3} \sin \theta \cos \theta \right) \left(\frac{\cos \theta}{\sin \theta} \right) v}$$

$$\Rightarrow \cos \theta = \frac{1}{2}$$

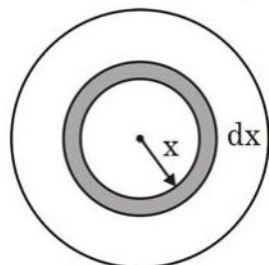
$$\Rightarrow \theta = 60^\circ$$

6. Ans (2)

$$E = \frac{x}{2} \frac{dB}{dt}$$

$$\Rightarrow E = \frac{3Kxt^2}{2}$$

$$\Rightarrow d\tau = (dq)Ex = \left(\frac{3Kxt^2}{2}\right) \left(\frac{2\pi x dx}{\pi r^2}\right) qx$$



$$\Rightarrow \tau = \frac{3Kt^2q}{r^2} \int_0^r x^3 dx$$

$$\Rightarrow \tau = \frac{3Kqt^2}{r} r^2 \dots\dots\dots(1)$$

torque due to friction force

$$d\tau = \mu g x dm = \mu g x (2\pi x dx) \frac{m}{\pi r^2}$$

$$\Rightarrow \tau = 2\mu g \frac{m}{r^2} \int_0^r x^2 dx = \frac{2}{3} \mu mgr$$

$$\Rightarrow \frac{3Kqt^2r^2}{4} = \frac{2}{3} \mu mgr$$

$$\Rightarrow t = \sqrt{\frac{8\mu mg}{9Kqr}}$$

Substituting values, we get

$$t = 2s$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (D)

On increasing volume concentration decreases.

4. Ans (B)

Q must be an ester. Addition of ethyl magnesium chloride to an alkyl acetate gives II, formate ester gives III and propionic ester gives IV. I can be produced from acetone and not from ester.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (C,D)

$$V = \frac{nRT}{P}$$

$$V_A = 200R, V_C = 400R, V_B = 400R$$

Obtaining equation for AB process

$$P = \frac{400R}{600R - V}$$

$$w_{AB} = - \int P dv$$

$$= - 400R \ln 2$$

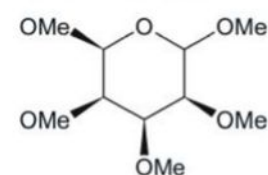
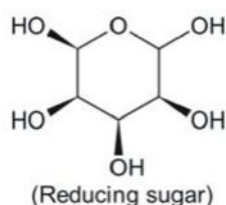
$$\Delta S = nC_V \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} = nC_P \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$$

$$= 0.2 \text{ L.atm}$$

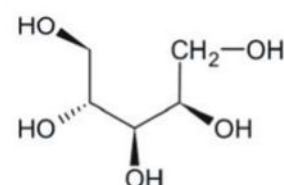
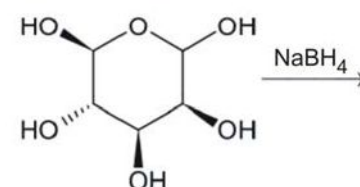
6. Ans (A,B,C)

Since anomeric carbon is a chiral carbon, so given sugar can exist in two anomeric pyranose forms.

Since it is in hemiacetal, so it can reduce to tollen's reagent to silver mirror.



(Non-reducing because anomeric -OH is not free)



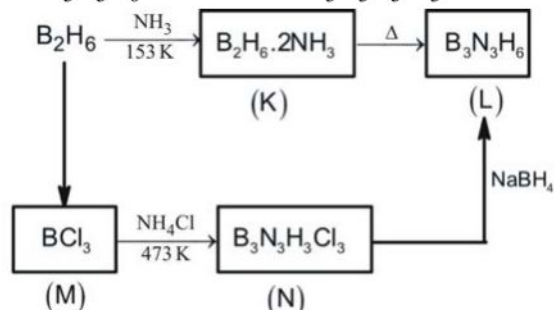
PART-2 : CHEMISTRY

SECTION-I (iii)

8. Ans (D)

K : $[\text{BH}_2(\text{NH}_3)_2]$ [BH₄] M: BCl₃

L : B₃N₃H₆ N: B₃N₃H₃Cl₃

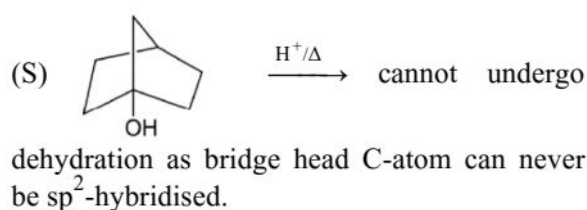
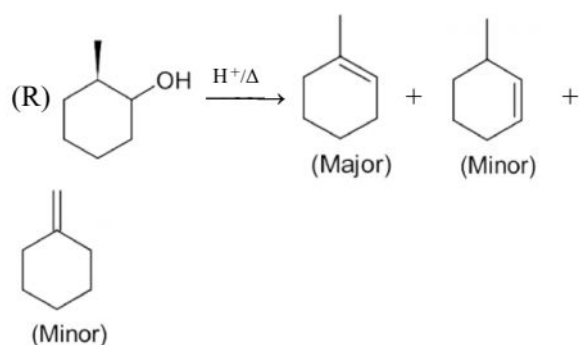
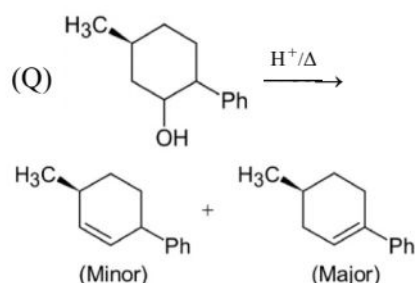
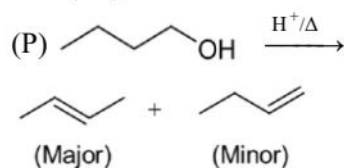


9. Ans (B)

$$1^{\text{st}} \text{ order: } t_{1/2} = \frac{0.693}{k}, k = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$3 : t_{1/2} = \frac{2}{k} \left(\sqrt{a} - \sqrt{\frac{a}{2}} \right)$$

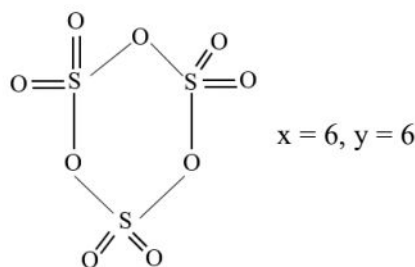
10. Ans (D)



PART-2 : CHEMISTRY

SECTION-II

2. Ans (12)



3. Ans (85)

Let KI solution is M molar so m.mol = $20 \times M$

$$\text{Given, } \text{KIO}_3: \frac{1}{10} M$$

$$\Rightarrow 20 \times M = \frac{1}{10} \times 30$$

$$\Rightarrow \text{Molarity of KI} = 0.15 \text{ M}$$

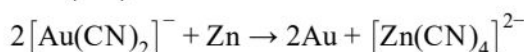


$$\text{m.mol: } 5 \quad 10$$

$$\Rightarrow \text{AgI} = 0.3 \times 50 - 10 = 5 \text{ m.mol}$$

$$\Rightarrow \% \text{AgNO}_3 = 85\%$$

4. Ans (20)



$$E^\circ = -0.60 - (-1.26) = 0.66 \text{ V}$$

Since E° value for the reduction of complexed Ag^+ ion to Ag is higher than that of complexed Au^+ ion into Au. So, Ag^+ ion will be reduced first.

$$\text{Mole of Zn added} = \frac{78}{65} = 1.2$$

$$\text{Moles of } [\text{Ag}(\text{CN})_2]^- = 0.003 \times 500 = 1.5$$

$$\text{Moles of } [\text{Au}(\text{CN})_2]^- = 0.002 \times 500 = 1.0$$

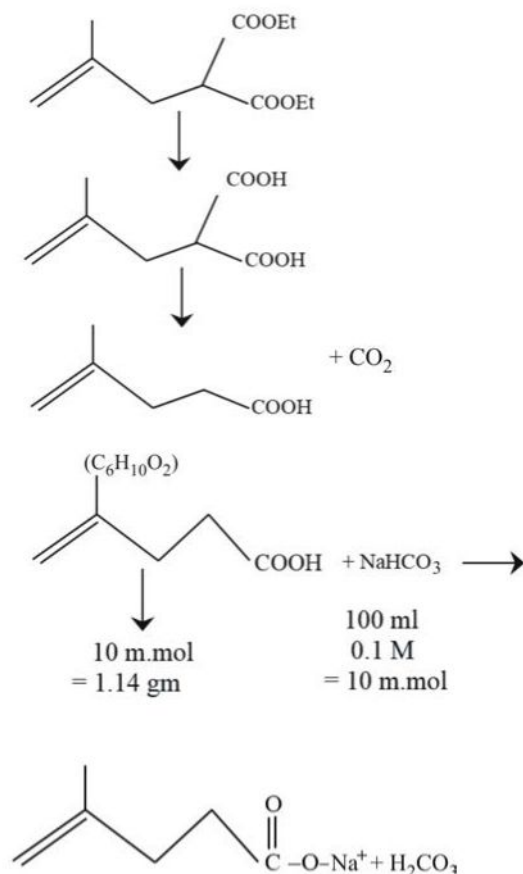
Moles of Zn used to reduce 1.5 moles of $[\text{Ag}(\text{CN})_2]^- = 0.75$

Moles of $[\text{Au}(\text{CN})_2]^-$ reduced by remaining 0.45 moles of Zn = 0.90

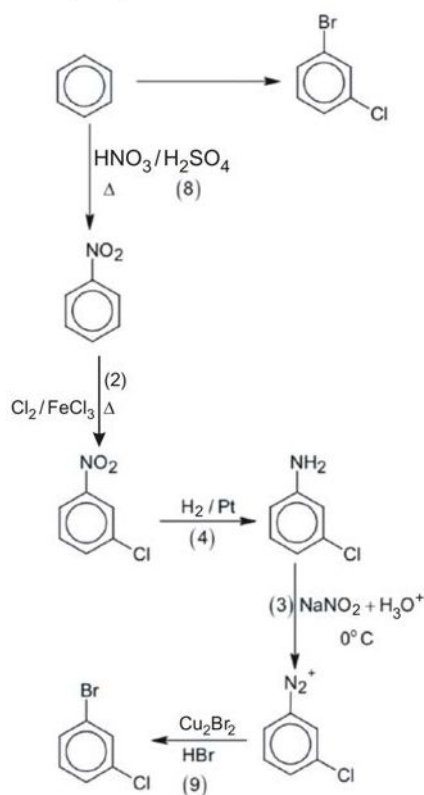
$$\text{Moles of } [\text{Au}(\text{CN})_2]^- \text{ left in the solution} = 0.10$$

$$\text{Concentration of } [\text{Au}(\text{CN})_2]^- \text{ left in the solution} = \frac{0.10}{500} = 2 \times 10^{-4} \text{ M} = x$$

5. Ans (1140)



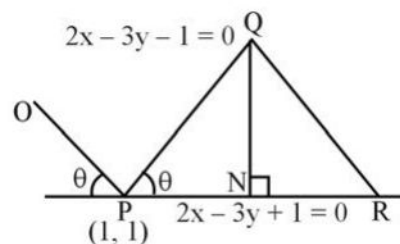
6. Ans (39)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)



$$\text{Let } Q = \left(t, \frac{2t-1}{3}\right)$$

$$PQ = QN \csc \theta$$

$$\Rightarrow t = \frac{47}{13}, -1$$

For $t = -1$, Q lies on OP produced which is not possible

$$\Rightarrow Q \equiv \left(\frac{47}{13}, \frac{27}{13}\right)$$

$$\text{Let } R = \left(\frac{3\alpha-1}{2}, \alpha\right)$$

$$\text{Slope of } QR = \text{slope of } OP \Rightarrow \alpha = 53/13$$

2. Ans (A)

$$S_1 < 0 \Rightarrow m^2 - 3m + 2 < 0 \Rightarrow m \in (1, 2)$$

$$(2m-1-2) \left(2 \left(\frac{3}{2}\right) - 0 - 2\right) < 0 \Rightarrow m < \frac{3}{2}$$

$$\text{hence, } m \in \left(1, \frac{3}{2}\right)$$

3. Ans (A)

$$h'(x) = (2x^2 - \ln x) g(x)$$

$$g'(x) = \frac{1}{\ln x^3} 3x^2 - \frac{1}{\ln x^2} 2x$$

$$g'(x) = \frac{x(x-1)}{\ln x} > 0 \forall x > 1; g(x) > g(1) \Rightarrow g(x) > 0 \forall x > 1$$

For $h(x)$ is increasing

$$h'(x) > 0 \Rightarrow 2x^2 - \ln x > 0 \text{ as } (g(x) > 0)$$

$$\text{Let } H(x) = 2x^2 - \ln x$$

$$H'(x) = 4x - \frac{1}{x} = \frac{4x^2 - 1}{x} > 0 \text{ where } x > 1$$

$$H(x) > H(1) \Rightarrow H(x) > 2$$

$$\therefore h'(x) > 0 \forall x \in (1, \infty)$$

$$\therefore h(x) \text{ is increasing on } (1, \infty)$$

4. Ans (B)

The given equation is $x^3 + y^3 = a^3$

Differentiating both sides w.r.t. x we have

$$\frac{dy}{dx} = \frac{-x^2}{y^2}$$

$$\left(\frac{dy}{dx}\right)_{(x', y')} = -\left(\frac{x'}{y'}\right)^2 \dots\dots(1)$$

Equation of the tangent at (x', y') is $y - y'$

$$= -\left(\frac{x'}{y'}\right)^2 (x - x')$$

$$(x')^2 x + (y')^2 y = (x')^3 + (y')^3$$

This passes through (x_1, y_1) . Therefore

$$(x')^2 x_1 + (y')^2 y_1 = (x')^3 + (y')^3 = a^3 = x_1^3 + y_1^3$$

As (x_1, y_1) lies on the curve. Hence

$$x_1 (x'^2 - x_1^2) = -y_1 (y'^2 - y_1^2)$$

$$-\frac{x_1 (x' + x_1)}{y_1 (y' + y_1)} = \frac{y' - y_1}{x' - x_1}$$

= Slope of the line joining (x', y') and (x_1, y_1)

Which is the slope of the tangent at (x', y') . So

from equation (i)

$$\frac{-x_1 (x' + x_1)}{y_1 (y' + y_1)} = -\left(\frac{x'}{y'}\right)^2$$

$$x_1 x' (y')^2 + x_1^2 (y')^2 = y_1 y' x'^2 + y_1^2 x'^2$$

$$x' y' (x_1 y' - x y_1) = -(x_1 y' - x y_1)(x_1 y' + x y_1) \dots$$

(ii)

Suppose $x_1 y' - x y_1 = 0$ so that

$$\frac{x'}{x_1} = \frac{y'}{y_1} = \lambda$$

Hence, $x' = \lambda x_1, y' = \lambda y_1$ so that

$$a^3 = x'^3 + y'^3 = \lambda^3 (x_1^3 + y_1^3) = \lambda^3 a^3$$

Which implies $\lambda = 1$ and hence $x' = x_1$ and y_1

= y' a contradiction. Therefore $x_1 y' - x y_1 \neq 0$

.Hence from equation (ii)

$$x' y' = -(x_1 y' + x y_1)$$

Dividing both sides with $x' y'$ we have

$$\frac{x_1}{x'} + \frac{y_1}{y'} = -1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C)

$$f(x) = \sin x + px + q$$

$$f'(x) = \cos x + p$$

If $p > 1$ then $f(x)$ is increasing. So only one real root, which is positive if $q < 0$ and negative if $q > 0$.

If $p < -1$ $f(x)$ is decreasing so only one real root, which is negative if $q < 0$

6. Ans (A,C)

$$\vec{u} \cdot \vec{w} = \cos \alpha \cos \gamma + \sin \alpha \sin \gamma = \cos (\alpha - \gamma)$$

$$\vec{v} \cdot \vec{w} = \cos (\beta - \gamma), \vec{u} \cdot \vec{v} = \cos (\alpha - \beta)$$

$$\text{Let } A = \alpha - \gamma, B = \beta - \gamma, \alpha - \beta = A - B$$

$$\text{Now, } \vec{u} \cdot \vec{w} = \vec{v} \cdot \vec{w}$$

$$\Rightarrow \cos A = \cos B \Rightarrow B = 2m\pi + A \text{ where } m \in \mathbb{I}$$

$$\text{But } -2\pi < A, B < 2\pi$$

$$\therefore B = A, B = -A, B = 2\pi - A \text{ or } B = -2\pi + A$$

$$\text{when } B = A \text{ or } B = -2\pi + A$$

$$\text{we get } \cos A = -\sqrt{3} \cos 0 \text{ or } -\sqrt{3} \cos(-2\pi)$$

$$\Rightarrow \cos A = -\sqrt{3}. \text{ Not possible}$$

$$\text{when } B = -A \text{ or } B = 2\pi - A, \text{ we get}$$

$$\cos A = -\sqrt{3} \cos(2A)$$

$$\Rightarrow 2\sqrt{3} \cos^2 A + \cos A - \sqrt{3} = 0$$

$$\Rightarrow \cos A = \frac{-1 \pm \sqrt{1 + 4(2\sqrt{3})}(\sqrt{3})}{2(2\sqrt{3})}$$

$$= \frac{-1 \pm 5}{4\sqrt{3}} = \frac{1}{\sqrt{3}}, \frac{\sqrt{3}}{2}$$

$$\text{Now, } \vec{u} \cdot \vec{v} = \cos(2A)$$

$$= 2\cos^2 A - 1 = \begin{cases} -\frac{1}{3} \text{ when } \cos A = \frac{1}{\sqrt{3}} \\ \frac{1}{2} \text{ when } \cos A = -\frac{\sqrt{3}}{2} \end{cases}$$

7. Ans (B,C,D)

$$3PQP^{-1} + 3P = 2P^{-1}QP + 2P$$

$$\Rightarrow 3PQP + 3PPP^{-1} = 2P^{-1}QP + 2P^{-1}PP$$

$$\Rightarrow |3P(Q+P)P^{-1}| = |2P^{-1}(Q+P)P|$$

$$\Rightarrow 3^n |P+Q| = 2^n |P+Q|$$

$$\Rightarrow |P+Q| = 0$$

$$(c) 3PQP^{-1} - 3P^{-1}QP = -P^{-1}QP - P$$

$$= -P^{-1}QP - P^{-1}PP$$

$$\Rightarrow 3^n |PQP^{-1} - P^{-1}QP| = (-1)^n |Q+P| = 0$$

PART-3 : MATHEMATICS

SECTION-I (iii)

8. Ans (B)

$$(P) \frac{dy}{dx} = \frac{-y \sin x}{1 + \cos x}$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{-\sin x}{1 + \cos x} dx$$

$$\Rightarrow \ln y = \ln(1 + \cos x) + \ln C$$

$$\Rightarrow y = (1 + \cos x)C$$

$$\because y\left(\frac{\pi}{2}\right) = -1$$

$$\therefore y = -(1 + \cos x) \Rightarrow y(0) = -2$$

$$(Q) \text{ At } x = \frac{-\pi}{2}, -1, 0, 1, \frac{\pi}{2}, f(x) \text{ is non-derivable}$$

$$(R) \text{ Since } e^{\ln(x+1)} = x+1, \text{ thus } e^{\ln(x+1)} = |y|$$

$$\Rightarrow x+1 \geq |y|$$

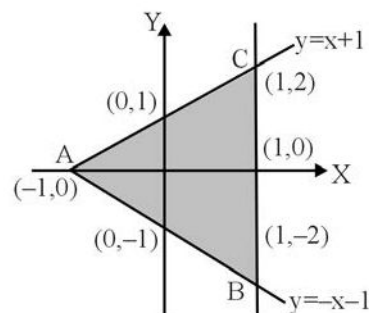
$$\Rightarrow x+1 \leq 0, x \geq 1 \text{ or } (x+1) \geq 0$$

$$\text{and } -(x+1) \leq y \leq (x+1).$$

Thus the region represented by inequalities $-(x+1)$

$< y < (x+1)$ and $|x| \leq 1, -1 \leq x \leq 1$ is as shown

below



Thus required area is the area of $\triangle ABC =$

$$\frac{1}{2}(4)(2) = 4 \text{ square units}$$

(S)

$$\frac{(\sec^{-1}x + \cos^{-1}x)^3 - 3\sec^{-1}x \cos^{-1}x (\sec^{-1}x + \cos^{-1}x)}{\left(\frac{\pi}{2}\right)^3} = 7$$

$$\Rightarrow \frac{\pi^3}{8} - 3\sec^{-1}x \cos^{-1}x \left(\frac{\pi}{2}\right) = \frac{7\pi^3}{8}$$

$$\Rightarrow \sec^{-1}x = \pi \text{ or } \sec^{-1}x = \frac{-\pi}{2}$$

$$\Rightarrow x = -1$$

9. Ans (D)

(P) $\sin A \sin 2A \sin 3A \sin 4A$

$$= ax^2 + bx^3 + cx^4 + dx^5$$

$$a + b + c + d = f(1)$$

Put $x = 1$

$$\Rightarrow \sin^2 A = 1$$

$$A = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow \sin A \cdot \sin 2A \cdot \sin 3A \cdot \sin 4A = 0$$

(Q) $2 \sin x \cos x = \frac{1}{\sqrt{2}}$

$$\Rightarrow \sin 2x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 2x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\Rightarrow x = \frac{\pi}{8}, \frac{3\pi}{8}$$

$$\Rightarrow d = \frac{3\pi}{8} - \frac{\pi}{8} = \frac{\pi}{4}$$

(R) $4 \cos x (2 - 3 \sin^2 x) + (\cos 2x + 1) = 0$

$$\Rightarrow 4 \cos x (-1 + 3 \cos^2 x) + 2 \cos^2 x = 0$$

$$\Rightarrow 2 \cos x [-2 + 6 \cos^2 x + \cos x] = 0$$

$$\Rightarrow 2 \cos x (2 \cos x - 1) (3 \cos x + 2) = 0$$

$$\Rightarrow \cos x = 0, \frac{1}{2}, \frac{-2}{3}$$

$$\text{In } x \in \left[0, \frac{\pi}{2}\right] \Rightarrow x = \frac{\pi}{2}, \frac{\pi}{3}$$

Least difference between the roots

$$= \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$$

(S) $f(x) = \sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x}$

$$= \sqrt{\sin^4 x + 4 - 4 \sin^2 x} - \sqrt{\cos^4 x + 4 - 4 \cos^2 x}$$

$$= 2 - \sin^2 x - (2 - \cos^2 x)$$

$$= \cos^2 x - \sin^2 x = \cos 2x$$

$$g(\sin 2t) = \sin t + \cos t$$

$$(g(\sin 2t))^2 = 1 + \sin 2t$$

$$\Rightarrow (g(\sin 2t))^2 = 1 + \sin 2t$$

$$\Rightarrow g(\sin 2t) = \sqrt{1 + \sin 2t}$$

$$\therefore g(x) = \sqrt{1+x}, -1 \leq x \leq 1$$

Now, $g\{f(x)\} = g(\cos 2x)$

$$= \sqrt{1 + \cos 2x} = \sqrt{2} |\cos x|$$

$$\therefore \text{Range of } g\{f(x)\} \text{ is } [0, \sqrt{2}]$$

$$\Rightarrow a^2 + b^2 = 2$$

10. Ans (A)

(P)

$$\lim_{h \rightarrow 0^+} 2 \int_0^1 \frac{h dx}{h^2 + x^2} = \lim_{h \rightarrow 0^+} \left[2 \frac{h}{h} \tan^{-1} \frac{x}{h} \right]_0^1 = \pi$$

(Q) Put $x - 2 = x, y + 1 = y$

The required region is defined by $1 \leq |x| + |y| \leq 2$

$$\text{Required area} = (2\sqrt{2})^2 - (\sqrt{2})^2 = 6$$

(R) $\int_0^{110\pi} [\sin x + \cos x] dx = 55 \int_0^{2\pi} [\sin x + \cos x] dx$

$$= 55 \int_{\frac{\pi}{4}}^{\frac{9\pi}{4}} [\sqrt{2} \sin \theta] d\theta$$

$$= -55\pi$$

(S) $\frac{\left(\frac{x-2y+3}{\sqrt{5}}\right)^2}{4} + \frac{\left(\frac{2x+y+1}{\sqrt{5}}\right)^2}{\frac{1}{4}} = 1$

$$a^2 = 4, b^2 = \frac{1}{4} \Rightarrow e = \frac{\sqrt{15}}{4}$$

$$k^2 = 4a^2 e^2 = 4 \times 4 \times \frac{15}{16} = 15$$

11. Ans (D)

(P) $9 - a^2 > 5$

$\Rightarrow -2 < a < 2 \Rightarrow$ number of integral values = 3

(Q) $b + c = a^3 - a$ & $bc = a^2$

$x^2 - (a^3 - a)a + a^2 = 0$ has two roots b & c

as b & c are real so

$D \geq 0$

$(a^3 - a)^2 - 4a^2 \geq 0$

$a^2 [(a^2 - 1)^2 - 4] \geq 0$

$a^2 [a^4 - 2a^2 - 3] \geq 0$

$a^2 (a^2 + 1)(a^2 - 3) \geq 0$

$a^2 - 3 \geq 0$

$a^2 \geq 3 \Rightarrow$ Least value of a^2 is 3.

(R) number of terms in $(a + b + c - d)^n$ is ${}^{n+3}C_3$

and the sum of coefficients = 2^n

So $2^n > {}^{n+3}C_3$ which is true for $n > 6$

(S) $\lim_{x \rightarrow \infty} (8x^3 + ax^2)^{\frac{1}{3}} - bx$

$= \lim_{x \rightarrow \infty} (2x) \left(1 + \frac{a}{8x} \right)^{\frac{1}{3}} - bx$

$= \lim_{x \rightarrow \infty} 2x \left(1 + \frac{a}{24x} \right) - bx$

$= \lim_{x \rightarrow \infty} (2 - b)x + \frac{a}{12}$

for limit to exist $2 - b = 0$

$\Rightarrow b = 2$ & $\frac{a}{12} = 1 \Rightarrow a = 12$

$a - 3b = 6$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1)

Put $x^2 = X \Rightarrow x dx = \frac{dX}{2}$

$y^2 = Y \Rightarrow y dy = \frac{dY}{2}$

$(2X + 3Y - 7) \frac{dX}{2} - (3X + 2Y - 8) \frac{dY}{2} = 0$

$\frac{dY}{dX} = \frac{2X + 3Y - 7}{3X + 2Y - 8}$

Reduce it to variable separable by substituting

$X = X' + h$

$Y = Y' + k$

We get the solution

$(x^2 + y^2 - 3) = (x^2 - y^2 - 1)^5 C$

2. Ans (6)

$\left| |z| - \left| \frac{5}{z} \right| \right| \leq \left| z - \frac{5}{z} \right| \Rightarrow \left| |z| - \frac{5}{|z|} \right| \leq 4$

Let $|z| = t$

$-4 \leq t - \frac{5}{t} \leq 4 \Rightarrow -4t \leq t^2 - 5 \leq 4t$

Now, $t^2 + 4t - 5 \geq 0$ $t^2 - 4t - 5 \leq 0$

$(t + 5)(t - 1) \geq 0$ $(t - 5)(t + 1) \leq 0$

$\Rightarrow t \in [1, \infty)$ $t \in [0, 5]$ (as t is positive)

$\therefore t \in [1, 5]$

3. Ans (9)

$\left(\frac{x^4 + x^2 + 1}{x^2 - x + 1} - \frac{x^2 + 1}{x} \right)^{12} = \left(x^2 - \frac{1}{x} \right)^{12}$

$\Rightarrow 9^{\text{th}}$ term is free from 'x'

4. Ans (31)

The key is to count backwards. First choose the pair which you pick on Wednesday in 5 ways. Then there are four pairs of socks for you to pick a pair of on Tuesday and don't want to pick a pair. Since there are 4 pairs, the number of ways to do this is $\binom{8}{2} - 4$. Then, there are two pairs and two nonmatching socks for you to pick from on Monday a total of 6 socks. Since you don't want to pick a pair, the number of ways to do this is $\binom{6}{2} - 2$. Thus the answer is

$$\frac{(5) \left(\binom{8}{2} - 4 \right) \left(\binom{6}{2} - 2 \right)}{\binom{10}{2} \binom{8}{2} \binom{6}{2}} = \frac{26}{315}$$

$$\frac{m+n}{11} = \frac{315+26}{11} = \frac{341}{11} = 31$$

5. Ans (8)

∵ Elements of determinant can be 1 or -1
 ∵ First two places of first row can be filled in $2 \times 2 = 4$ ways and last place can be filled in only one way to product be 1
 ∵ First row can be made in 4 ways
 Similarly, second row can be made in 4 ways and third row can be made in 1 ways only
 ∵ $m = 16$

$$\therefore |P| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = 1$$

$$\therefore \text{adj}(\text{adj} P) = |P|^{n-2} \cdot P = P \Rightarrow \text{tr}(\text{adj}(\text{adj} P)) = \text{tr}(P) = 2 = n$$

$$\therefore \frac{m}{n} = \frac{16}{2} = 8$$

6. Ans (23)

Let the position vectors of A, B and C be $\vec{O}$, $\vec{b}$ and $\vec{c}$ -respectively.

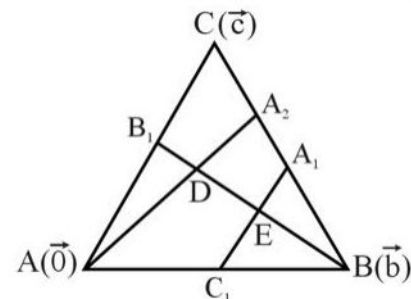
$$\text{We have, } \vec{AC}_1 = \frac{\vec{b}}{2}, \vec{AB}_1 = \frac{\vec{c}}{2}, \vec{AA_1} = \frac{\vec{b} + \vec{c}}{2}, \vec{AA_1} = \frac{3\vec{c} + \vec{b}}{4}$$

Equation of lines BB_1 , AA_2 and C_1A_1 are respectively

$$\vec{r} = \vec{b} + \lambda_1 \left(\frac{\vec{c}}{2} - \vec{b} \right)$$

$$\vec{r} = \lambda_2 \frac{3\vec{c} + \vec{b}}{4} \text{ and } \vec{r} = \frac{\vec{b}}{2} + \lambda_3 \left(\frac{\vec{c}}{2} \right)$$

$$\text{For point D, we have } \vec{b} + \lambda_1 \left(\frac{\vec{c}}{2} - \vec{b} \right) = \lambda_2 \left(\frac{3\vec{c} + \vec{b}}{4} \right)$$



$$\Rightarrow \vec{b} \left(1 - \lambda_1 - \frac{\lambda_2}{4} \right) + \frac{\vec{c}}{4} (2\lambda_1 - 3\lambda_2) = \vec{0}$$

$$\Rightarrow \lambda_1 = \frac{6}{7}, \lambda_2 = \frac{4}{7}$$

$$\Rightarrow \vec{AD} = \frac{3\vec{c} + \vec{b}}{7}$$

$$\text{For point E we have } \vec{b} + \lambda_1 \left(\frac{\vec{c}}{2} - \vec{b} \right) = \frac{\vec{b}}{2} + \frac{\lambda_3}{2} \vec{c}$$

$$\Rightarrow \lambda_1 = \lambda_3 = \frac{1}{2} \Rightarrow \frac{2\vec{b} + \vec{c}}{4}$$

$$\text{Now, } \vec{EA_1} = \frac{3\vec{c} + \vec{b} - 2\vec{b} - \vec{b}}{4} = \frac{2\vec{c} - \vec{b}}{4},$$

$$\vec{DA_1} = \frac{\vec{b} + \vec{c}}{2} - \frac{3\vec{c} + \vec{b}}{7} = \frac{5\vec{b} + \vec{c}}{14}$$

$$\text{Area of the quadrilateral } EA_1A_2D = \frac{1}{2} |\vec{EA_2} \times \vec{DA_1}|$$

$$= \frac{1}{112} |(2\vec{c} - \vec{b}) \times (5\vec{b} + \vec{c})| = \frac{1}{112} |10\vec{c} \times \vec{b} - \vec{b} \times \vec{c}|$$

$$= \frac{11}{112} |\vec{c} \times \vec{b}| = \frac{11}{56} \times \frac{1}{2} |\vec{c} \times \vec{b}| = \frac{11}{56} \text{ (area of } \triangle ABC)$$

$$\text{Thus required ratio of is } \frac{11}{56}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

19-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	A	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	A,C,D	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	0.25	0.00	1.41	6.70 to 6.85		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	6	2	5	6	2	6

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C	A,C,D	B,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	8.00	6.00	130.00	365.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	5625	9	4	3	3	3

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	A	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	A,B,C,D	A,B,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.00	3.00	18.00	3.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	3	5	1	80	2	4

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

According to Newton's second Law, we have

$$F = \frac{\Delta p}{\Delta t} = \left(\frac{\Delta m}{\Delta t} \right) (\Delta v)$$

$$\Rightarrow F = \rho \left(\frac{\Delta V}{\Delta t} \right) (2v)$$

$$\Rightarrow F = \rho(Av)(2v) = 2\rho Av^2$$

2. Ans (A)

$$A = A_0 e^{-\lambda t} \Rightarrow \frac{6}{100} = e^{-\frac{\ln 2}{T_1} t}$$

$$\frac{1}{\frac{t}{T_1}} = \frac{6}{100} \Rightarrow \frac{1}{2^4}$$

$$T_1 = 30 \text{ min}$$

3. Ans (C)

$$\frac{dA}{dt} = \frac{L}{2m}$$

4. **Ans (B)**

B_w is magnetic field of wire

$$B_1 = B_0 + B_w$$

$$B_2 = B_0 - B_w$$

$$\Rightarrow B_0 = \frac{B_1 + B_2}{2}$$

PART-1 : PHYSICS

SECTION-I (ii)

6. **Ans (A,C,D)**

$$(A) U = \frac{1}{2} CV^2$$

As source between the plates is connected, so potential difference remains constant. But capacitance C becomes KC hence energy stored is increased by factor K .

(B) electric field $\frac{V}{d}$ is not changed.

(C) Charge on each plate is increased by factor K hence force between them increases by a factor of K^2 .

(D) Since $Q = CV$

Hence charge becomes KQ as C becomes KC and V remains unchanged.

7. **Ans (A,B,C,D)**

Let ℓ_1 and ℓ_2 be the length of the air column for the first and the second resonance respectively with a tuning fork of frequency f .

$$\text{Then } \lambda = 2(\ell_2 - \ell_1)$$

From relation, $v = f\lambda$

$$v = 2f(\ell_2 - \ell_1)$$

$$= 2 \times 480 \times (51 \text{ cm} - 16 \text{ cm}) = 336 \text{ m/s}$$

If T be the temperature of the air in air column of the resonance tube.

Then from formula,

$$\frac{v}{v_a} = \sqrt{\frac{T}{T_0}}$$

$$T = T_0 \left(\frac{v}{v_0} \right)^2 = 273 \left(\frac{336}{330} \right)^2 = 283\text{K}$$

$$\text{Also } \frac{\lambda}{4} = \ell_1 + e = 16 \text{ cm} + 0.3 \text{ d}$$

$$= \frac{v}{4f} = \frac{336}{4 \times 480} \text{ m}$$

Therefore $d = 5 \text{ cm}$

Again

$$\frac{\Delta v}{v} \times 100\% = \frac{\Delta \ell_2 + \Delta \ell_1}{\ell_2 - \ell_1} \times 100\% = \frac{4}{7}\%$$

And

$$\frac{\Delta T}{T} \times 100\% = 2 \times \frac{\Delta \ell_2 + \Delta \ell_1}{\ell_2 - \ell_1} \times 100\% = \frac{8}{7}\%$$

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (0.25)**

Velocity of particle undergoing wave motion is given by

$$\frac{\partial y}{\partial t} = -u \frac{\partial y}{\partial x}$$

At $t = 0$ and $x = 2$, we have

$$\frac{\partial y}{\partial t} = \frac{-1 \times 10 \times 10^{-3}}{4 \times 10^{-2}} = -\frac{1}{4} = -0.25 \text{ ms}^{-1}$$

2. **Ans (0.00)**

Displacement and velocity of particle is vector sum of displacements and velocity due to individual pulses.

So, displacement at $x = 8 \text{ cm}$ and $t = 0.06 \text{ sec}$ is $y = 5 \text{ mm} - 5 \text{ mm} = 0$

3. **Ans (1.41)**

AT an instant when m is just leaving M , let v_1 and v_2 be the velocities of m and M (both horizontal).



By law of conservation of linear momentum, we have

$$1 \times v_1 = 4v_2 \dots\dots\dots(1)$$

By law of conservation of Mechanical energy, we have

$$(1)(10)(4 - 2) = \frac{1}{2}(1)v_1^2 + \frac{1}{2}(4)v_2^2$$

$$\Rightarrow v_1^2 + 4v_2^2 = 40 \dots\dots\dots(2)$$

Solving these two equations, we get

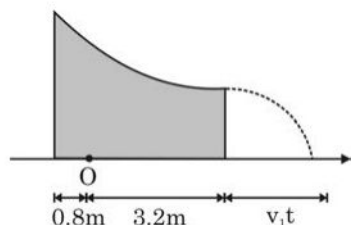
$$v_2 = \sqrt{2} \text{ ms}^{-1} \text{ and } v_1 = 4\sqrt{2} \text{ ms}^{-1}$$

4. **Ans (6.70 to 6.85)**

$$\sum m_g x_L = \sum m_L x_L$$

$$\Rightarrow 4d = 1(4 - d)$$

$$\Rightarrow d = 0.8 \text{ m}$$



i.e., m is already at $x = 3.2 \text{ m}$ at this instant.

Time taken by m to fall to ground from this instant will be

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 2}{10}} = \sqrt{\frac{2}{5}} \text{ s}$$

The x -coordinate of m where it strikes the ground will be

$$x = 3.2 + v_1 t = 3.2 + 4\sqrt{2} \times \sqrt{\frac{2}{5}} = 6.8 \text{ m}$$

PART-1 : PHYSICS

SECTION-II (ii)

5. **Ans (6)**

$$Q - \text{factor} = \frac{\omega_0}{2\Delta\omega}$$

6. **Ans (2)**

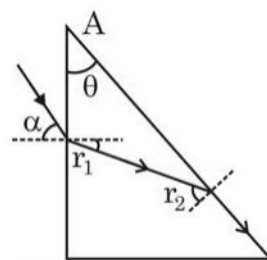
$$\Delta P = \sigma \left(\frac{1}{D/2} + \frac{1}{\infty} \right) = \frac{2\sigma}{D}$$

$$\text{Net force} = \sigma \pi D - (\Delta P) \text{ area}$$

$$= \sigma \pi D - \frac{2\sigma}{D} \frac{\pi D^2}{4} = \frac{\sigma \pi D}{2}$$

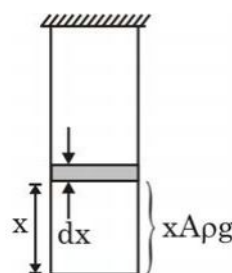
7. **Ans (5)**

$$\text{Angle of prism, } A = r_2 - r_1$$



8. **Ans (6)**

Consider an element as shown in the figure.



$$\text{Stress in the element} = \frac{\text{Force}}{\text{Area}} = \frac{x A p g}{A} = x \rho g$$

Now, elastic potential energy stored in the wire is

$$dU = \frac{1}{2} (\text{Stress}) (\text{Strain}) (\text{Volume})$$

$$\Rightarrow dU = \frac{1}{2} \frac{(\text{Stress})^2}{Y} (\text{Volume})$$

$$\Rightarrow dU = \frac{1}{2} \frac{(x \rho g)^2}{Y} A dx = \frac{1}{2} \frac{\rho^2 g^2 A}{Y} x^2 dx$$

Total elastic potential energy is

$$U = \frac{1}{2} \frac{\rho^2 g^2 A}{Y} \int_0^L x^2 dx$$

$$U = \frac{\rho^2 g^2 A L^3}{6Y}$$

$$\Rightarrow p = 6$$

10. **Ans (6)**

For equilibrium of cylinder in horizontal direction, we have

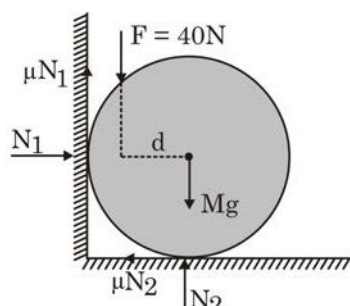
$$N_1 = \mu N_2 \dots\dots\dots(1)$$

For equilibrium of cylinder in vertical direction, we have

$$N_2 + \mu N_1 = F + Mg \dots\dots\dots(2)$$

Solving these two equations with $F = 40 \text{ N}$, $M = 2 \text{ kg}$ and $\mu = \frac{1}{3}$, we get

$$N_1 = 18 \text{ N and } N_2 = 54 \text{ N}$$



Since, $Fd = \mu(N_1 + N_2)r$

$$\Rightarrow d = \frac{\mu(N_1 + N_2)r}{F} = \frac{\left(\frac{1}{3}\right)(18 + 54)(0.1)}{40}$$

$$\Rightarrow d = 0.06 \text{ m} = 6 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B)**

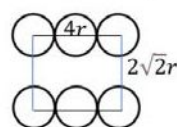
Plane P has the following arrangement of circular atoms:



Considering the smallest possible triangle in this arrangement

$$PF = \frac{\frac{1}{6} \times 3 \times \frac{22}{7} r^2}{\frac{1}{2} \times 2r \times \sqrt{3}r} = 0.907$$

And Plane Q has the arrangement:



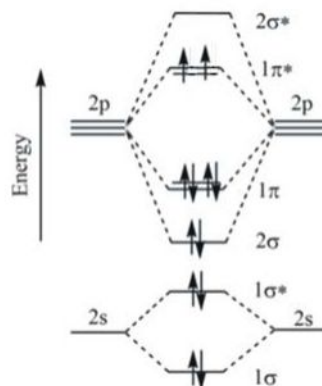
$$PF = \frac{\left[\frac{1}{2} \times 2 + \frac{1}{4} \times 4\right] \times \frac{22}{7} r^2}{4r \times 2\sqrt{2}r} = 0.5555$$

3. **Ans (D)**

Friedel-Craft acylation occur in the ring without deactivating $-\text{Cl}$ group

4. **Ans (C)**

$\text{O}_2 (16 e^-)$



Naming of molecular orbitals are as per preference of formation of σ & π bonds respectively.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,B,C)**

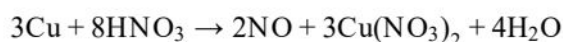
$$(A) z > 1 \frac{d_i}{d_r} > 1 \Rightarrow d_r < d_i$$

(B) Always repulsions $z > 1$

(C) First law of thermodynamics is valid for real gas (or) ideal gas

(D) 'z' linearly varies with p.

6. **Ans (A,C,D)**



$\text{NO} \Rightarrow$ has 11 valence e^- with an unpaired

e^- in $\pi^* 2p$ orbital

$\Rightarrow$ is diamagnetic in solid state due to dimerisation

$\Rightarrow$ it is neutral oxide

PART-2 : CHEMISTRY

SECTION-II (i)

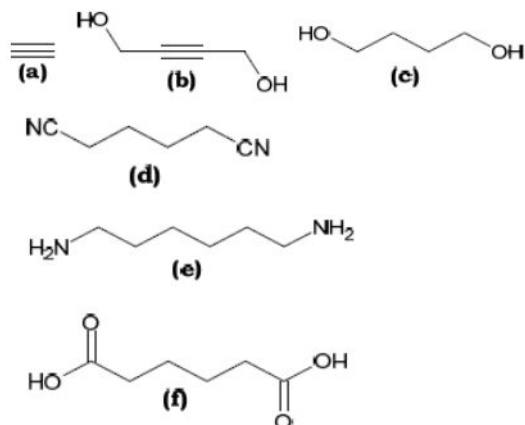
1. **Ans (8.00)**

$$\Delta G = [2 \times (-14) + (-4)] - [2 \times (-16) + 8] = -8 \text{ KJ/mol.}$$

2. **Ans (6.00)**

$$\Delta G = 3 \times (-14) - 3 \times (-16) = +6 \text{ KJ/mol.}$$

3. Ans (130.00)



Number of moles of hexamethylene diamine (e)

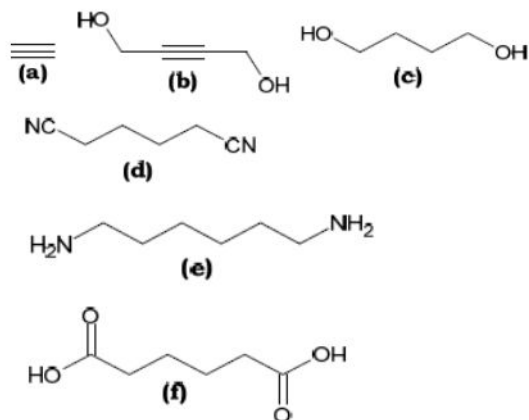
$$= \frac{290}{116} = 2.5$$

Number of moles of adipic acid (f) required = 2.5

Mass of adipic acid = $2.5 \times 146 \times 365$

Number of moles of acetylene (a) required = 5

4. Ans (365.00)



Number of moles of hexamethylene diamine (e)

$$= \frac{290}{116} = 2.5$$

Number of moles of adipic acid (f) required = 2.5

Mass of adipic acid = $2.5 \times 146 \times 365$

Number of moles of acetylene (a) required = 5

PART-2 : CHEMISTRY

SECTION-II (ii)

5. Ans (5625)

$$\frac{\ell}{A} = \frac{2}{4}$$

$$R = \rho \cdot \frac{\ell}{A} \Rightarrow 100 = \rho \cdot \frac{1}{2}$$

$$k = \frac{1}{200} \Omega^{-1} \text{ cm}^{-1}$$

$$\Lambda = \frac{1}{200} \times \frac{1000}{0.1} = 50$$

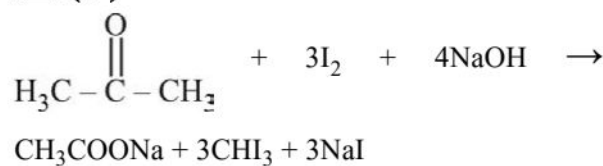
$$\Lambda^\circ = 300 + 100 = 400$$

$$\alpha = \frac{\Lambda}{\Lambda^\circ} = \frac{50}{400} = \frac{1}{8}$$

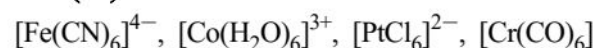
$$\Delta T_b = i \times K_b \times m$$

$$= \left(1 + \frac{1}{8}\right) \times 0.5 \times 0.1 = 0.05625$$

6. Ans (9)



7. Ans (4)



8. Ans (3)

Correct : 1, 3 & 7

$\Rightarrow$ number of correct statements = 3

Incorrect : remaining 4

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$\text{Put } x = 0, f(0) \cdot e^{f(0)} = 0$$

$$\Rightarrow f(0) = 0$$

$$\Rightarrow f \text{ is bijective so } f(x) > 0 \forall x \in (0, \infty)$$

$$f(x) e^{f(x)} = x$$

$$\ln f(x) + f(x) = \ln x$$

$$\frac{f(x)}{\ln x} = 1 - \frac{\ln f(x)}{\ln x} \dots (1)$$

$$\text{We know that } e^x > x$$

$$x \rightarrow f(x)$$

$$e^{f(x)} > f(x)$$

$$f(x) e^{f(x)} > f(x) \cdot f(x)$$

$$x > f^2(x)$$

$$\sqrt{x} > f(x)$$

$$\frac{\sqrt{x}}{x} > \frac{f(x)}{x} \Rightarrow 0 < \frac{f(x)}{x} < \frac{1}{\sqrt{x}}$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = 0$$

$$\Rightarrow \text{growth of } f < \text{growth of } x$$

$$\Rightarrow \text{growth of } \ln f(x) < \text{growth of } \ln x$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{\ln f(x)}{\ln x} = 0$$

$$\text{Putting this value in equation (1)}$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{\ln x} = 1$$

2. Ans (C)

$$x + xy + xyz = 1 \Rightarrow x(1 + y) = 1 - xyz \dots (1)$$

$$y + yz + xyz = 2 \Rightarrow y(1 + z) = 2 - xyz \dots (2)$$

$$z + zx + xyz = 4 \Rightarrow z(1 + x) = 4 - xyz \dots (3)$$

Multiply all above equation

$$xyz (1 + x)(1 + y)(1 + z) = (1 - xyz)(2 - xyz)(4 - xyz)$$

Let

$$xyz = p, (1 + x)(1 + y)(1 + z) = q$$

$$pq = (1 - p)(2 - p)(4 - p)$$

$$\text{Now } q = 1 + x + y + z + xy + yz + zx + xyz$$

$$q = 1 + x(1 + y) + y(1 + z) + z(1 + x) + p$$

$$q = 1 + 1 - p + 2 - p + 4 - p + p$$

$$q = 8 - 2p$$

$$p(8 - 2p) = (1 - p)(2 - p)(4 - p)$$

$$(4 - p)(2p - (1 - p)(2 - p)) = 0$$

$$(4 - p)(p^2 - 5p + 2) = 0$$

$$p = 4, \frac{5 + \sqrt{17}}{2}, \frac{5 - \sqrt{17}}{2}$$

$$\text{Largest value of } xyz = \frac{5 + \sqrt{17}}{2}$$

$$a = 5, b = 17, c = 2$$

$$a + b + c = 24$$

3. Ans (A)

$$y(x + y)^2 = x$$

d. w. r. to x

$$2y(x + y)(1 + y') + (x + y)^2 y' = 1$$

$$\Rightarrow y'(2y(x + y) + (x + y)^2) = 1 - 2y(x + y)$$

$$\Rightarrow (x + y) y' (2y + x + y) = 1 - 2y(x + y)$$

$$\Rightarrow (x + y) y' (x + 3y) + (x + y)(x + 3y)$$

$$= 1 - 2y(x + y) + (x + y)(x + 3y)$$

$$\Rightarrow (x + 3y)(x + y)(y' + 1)$$

$$= 1 + (x + y)(x + 3y - 2y)$$

$$\Rightarrow (x + 3y)((x + y)y' + 1) = 1 + (x + y)^2$$

$$\int \frac{1}{x + 3y} dx = \int \frac{(x + y)(y' + 1)}{1 + (x + y)^2} dx$$

$$\int \frac{1}{x + 3y} dy = \frac{1}{2} \ln((x + y)^2 + 1) + C$$

4. **Ans (D)**

From given condition odd elements of the domain can associate with unequal odd elements of the co-domain and even elements of the domain can associate with unequal even number of the co-domain.

Number of ways if x is odd

$$= 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 9$$

Number of ways if x is even

$$= 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 9$$

Total number of such functions = $9 \times 9 = 81$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**

$$f(x) = x^3 + 5x + 1$$

$$g'(y) = \frac{1}{f'(x)} = \frac{1}{3x^2 + 5}$$

when $y = 1$ then $x = 0$

$$\therefore g'(1) = \frac{1}{5}$$

$$h(g(g(x))) = x$$

$$\therefore x \rightarrow f(x)$$

$$h(g(x)) = f(x)$$

$$\therefore h(x) = f(f(x))$$

$$h(0) = f(f(0)) = 7$$

$$h(g(2))$$

$$f(f(g(2))) = f(2) = 8 + 10 + 1 = 19$$

$$h'(x) = f'(f(x)) \cdot f'(x)$$

$$h'(1) = f'(f(1)) \cdot f'(1)$$

$$= f'(7) \cdot f'(1)$$

$$= (147 + 5)(3 + 5) = 152 \times 8 = 1216$$

6. **Ans (A,B,C,D)**

$$X = {}^{20}C_3 = 1140$$

$$Y = \frac{{}^{20}C_1 \cdot {}^{16}C_2}{3} = 800$$

$$Z = {}^{20}C_3 - 10 \times {}^{18}C_1 = 1140 - 180 = 960$$

$$Y \cap Z = Y - 10 \times 14$$

$$= 800 - 140$$

$$= 660$$

7. **Ans (A,B,C)**

$$\therefore a(1 - e) = 5 \Rightarrow a = 20$$

$$\therefore b^2 = 400 \left(1 - \frac{9}{16} \right) = 175$$

$$\therefore P_1 P_2 = b^2 \therefore P_1 \times 5 = 175 \Rightarrow P_1 = 35$$

also centre of ellipse is $(-11, -8)$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (2.00)**

Since $g(x)$ has minimum value -1 at $x = a$

$$\text{therefore } g(a) = \int_0^a f(t) dt = -1 \text{ and}$$

$$g'(x) = f(a) = 0 \dots (1)$$

Now since $f(x)$ is increasing function in $[0, 1]$

$$\Rightarrow f(x) < 0 \text{ for } x \in [0, a) \text{ and } f(x) > 0 \text{ for } x \in (a, 1]$$

$$\Rightarrow f(x) + |f(x)| = \begin{cases} 0 & 0 \leq x < a \\ 2f(x) & a \leq x \leq 1 \end{cases}$$

$$\Rightarrow \int_0^1 f(t) + |f(t)| dt = \int_0^a 0 \cdot dt + \int_a^1 2f(t) dt = 6 \text{ (given)}$$

$$\Rightarrow \int_a^1 f(t) dt = 3$$

2. Ans (3.00)

Since $g(x)$ has minimum value -1 at $x = a$

$$\text{therefore } g(a) = \int_0^a f(t) dt = -1 \text{ and}$$

$$g'(x) = f(a) = 0 \dots (1)$$

Now since $f(x)$ is increasing function in $[0, 1]$

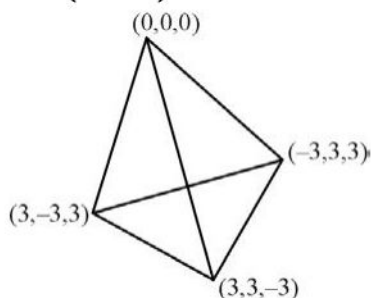
$$\Rightarrow f(x) < 0 \text{ for } x \in [0, a) \text{ and } f(x) > 0 \text{ for } x \in (a, 1]$$

$$\Rightarrow f(x) + |f(x)| = \begin{cases} 0 & 0 \leq x < a \\ 2f(x) & a \leq x \leq 1 \end{cases}$$

$$\begin{aligned} \text{Also } g(1) &= \int_0^1 f(t) dt \\ &= \int_0^a f(t) dt + \int_a^1 f(t) dt = -1 + 3 = 2 \end{aligned}$$

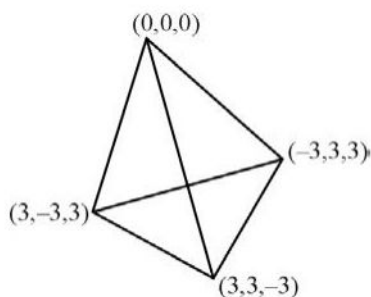
$$\begin{aligned} \text{Now } \int_0^1 g(x) f(x) dx &= \int_0^1 g(x) g'(x) dx = \left[\frac{g^2(x)}{2} \right]_0^1 \\ &= \frac{g^2(1) - g^2(0)}{2} = \frac{2^2 - 0}{2} = 2 \end{aligned}$$

3. Ans (18.00)



$$V = \left| \begin{vmatrix} 3 & -3 & 3 \\ -3 & 3 & 3 \\ 3 & 3 & -3 \end{vmatrix} \right| = \left| \begin{vmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \right| = 18$$

4. Ans (3.00)



$$\text{Distance} = \left| \frac{3}{\sqrt{3}} \right| = \sqrt{3}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (3)

$$f(x) > g(x)$$

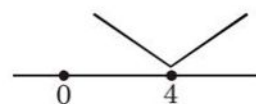
$$|x| + \frac{4}{|x|} > p|x| + 2 \quad \forall x \in \mathbb{R} - \{0\}$$

$$p < \frac{|x| + \frac{4}{|x|} - 2}{|x|}$$

$$p < 1 + \frac{4}{(x)^2} - \frac{2}{|x|} \quad \forall x \in \mathbb{R} - \{0\}$$

$$\text{Consider } h(x) = 1 + \frac{4}{x^2} - \frac{2}{x}, \quad x > 0$$

$$h'(x) = \frac{2(x-4)}{x^3}$$



$$\text{Now, } h(4) = 1 + \frac{1}{4} - \frac{1}{2} = \frac{3}{4}$$

$$p < h(x)$$

$$\therefore 0 < p < \frac{3}{4}$$

$$a = 0, \quad b = 3/4$$

$$\therefore a + 4b = 3$$

6. Ans (5)

$$P(E_i) = \frac{1}{2}, \quad P(E_m) = {}^{10}C_m \left(\frac{1}{2} \right)^{10}$$

$$P(E_i \cap E_m) = \frac{1}{2}, \quad {}^9C_{m-1} \left(\frac{1}{2} \right)^9 = \frac{{}^9C_{m-1}}{2^{10}}$$

$\therefore E_i$ and E_m are independent

$$\Rightarrow P(E_i \cap E_m) = P(E_i) \cdot P(E_m)$$

$$\Rightarrow \frac{{}^9C_{m-1}}{2^{10}} = {}^{10}C_m \left(\frac{1}{2} \right)^{10} \cdot \frac{1}{2}$$

$$\Rightarrow m = 5$$

7. Ans (1)

Let point P on the line is $(2\lambda + 1, -\lambda - 1, \lambda)$

The foot of perpendicular Q is given by

$$\frac{x - 2\lambda - 1}{1} = \frac{y + \lambda + 1}{1} = \frac{z - \lambda}{1} = \frac{-(2\lambda - 3)}{3}$$

$\therefore$ Q Lies on $x + y + z = 3$ and $x - y + z = 3$

$$\Rightarrow x + z = 3 \text{ and } y = 0$$

$$y = 0 \Rightarrow \lambda + 1 = \frac{-2\lambda + 3}{3} \Rightarrow \lambda = 0$$

$$\Rightarrow Q \text{ is } (2, 0, 1)$$

8. Ans (80)

$$\vec{a}_1 = (a, b), \vec{a}_2 = (a - 2, b - \sqrt{5}), \vec{a}_3 = (a + 2, b - \sqrt{5})$$

Area of Δ , whose vertices are $\vec{a}_1, \vec{a}_2$ and $\vec{a}_3$ is

$$\Delta = \frac{1}{2} \begin{vmatrix} 1 & a & b \\ 1 & a - 2 & b - \sqrt{5} \\ 1 & a + 2 & b - \sqrt{5} \end{vmatrix}$$

$$\Rightarrow \Delta^2 = \frac{1}{4} \begin{vmatrix} 1 & a & b \\ 1 & a - 2 & b - \sqrt{5} \\ 1 & a + 2 & b - \sqrt{5} \end{vmatrix} \begin{vmatrix} 1 & a & b \\ 1 & a - 2 & b - \sqrt{5} \\ 1 & a + 2 & b - \sqrt{5} \end{vmatrix}$$

and

$$|\vec{a}_1 - \vec{a}_2| = 3, |\vec{a}_1 - \vec{a}_3| = 3, |\vec{a}_2 - \vec{a}_3| = 4$$

$$\Rightarrow \Delta = \sqrt{5 \times 2 \times 2 \times 1} = 2\sqrt{5}$$

$$(2\sqrt{5})^2 = \frac{1}{4} \begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$$

$$\therefore \text{Required answer} = 4 \times 5 \times 4 = 80$$

9. Ans (2)

$$x - 4 = t + 1 \Rightarrow x - 5 = t$$

$$\therefore \sum_{r=0}^{3n} a_r (1+t)^r = \sum_{r=0}^{3n} A_r t^r$$

$$a_0 (1+t)^0 + a_1 (1+t)^1 + \dots + a_{2n} (1+t)^{2n} +$$

$$a_{2n+1} (1+t)^{2n+1} + \dots + a_{3n} (1+t)^{3n}$$

$$= A_0 t^0 + A_1 t^1 + \dots + A_{2n} t^{2n} + \dots + A_{3n} t^{3n}$$

compare coefficient of t^{2n} on both sides

$$A_{2n} = a_{2n} \cdot {}^{2n}C_{2n} + a_{2n+1} {}^{2n+1}C_{2n} + \dots + a_{3n} {}^{3n}C_{2n}$$

$$= {}^{2n}C_{2n} + {}^{2n+1}C_{2n} + {}^{2n+2}C_{2n} + \dots + {}^{3n}C_{2n}$$

$$[\because a_{2n} = a_{2n+1} = \dots = a_{3n} = 1]$$

$$= ({}^{2n+1}C_{2n+1} + {}^{2n+1}C_{2n}) + \dots + {}^{3n}C_{2n}$$

$$A_{2n} = {}^{3n+1}C_{2n+1}$$

$$\text{Similarly } B_{2n} = {}^{3n+1}C_{2n+1}$$

$$D_{2n} = {}^{3n+1}C_{2n+1}$$

$$\therefore \frac{A_{2n} + D_{2n}}{B_{2n}} = 2$$

10. Ans (4)

$$\sqrt{3}(\vec{a} \cdot \vec{b}) + |\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}| = 1 + |\vec{b}|^2$$

$$\sqrt{3}(\vec{a} \cdot \vec{b}) + |\vec{a} \times (\vec{b} \times \vec{a})| = 1 + |\vec{b}|^2$$

$$\sqrt{3}(\vec{a} \cdot \vec{b}) + |\vec{a}| |\vec{b} \times \vec{a}| \sin 90^\circ = 1 + |\vec{b}|^2$$

$$\sqrt{3}(\vec{a} \cdot \vec{b}) + |\vec{a} \times \vec{b}| = 1 + |\vec{b}|^2$$

$$\sqrt{3} |\vec{b}| \cos \theta + |\vec{b}| \sin \theta = 1 + |\vec{b}|^2$$

$$\frac{\sqrt{3} \cos \theta + \sin \theta}{\leq 2} = \frac{|\vec{b}| + \frac{1}{|\vec{b}|}}{\geq 2}$$

$$\therefore \theta = \frac{\pi}{6} \quad \therefore \operatorname{cosec}^2 \theta = 4$$